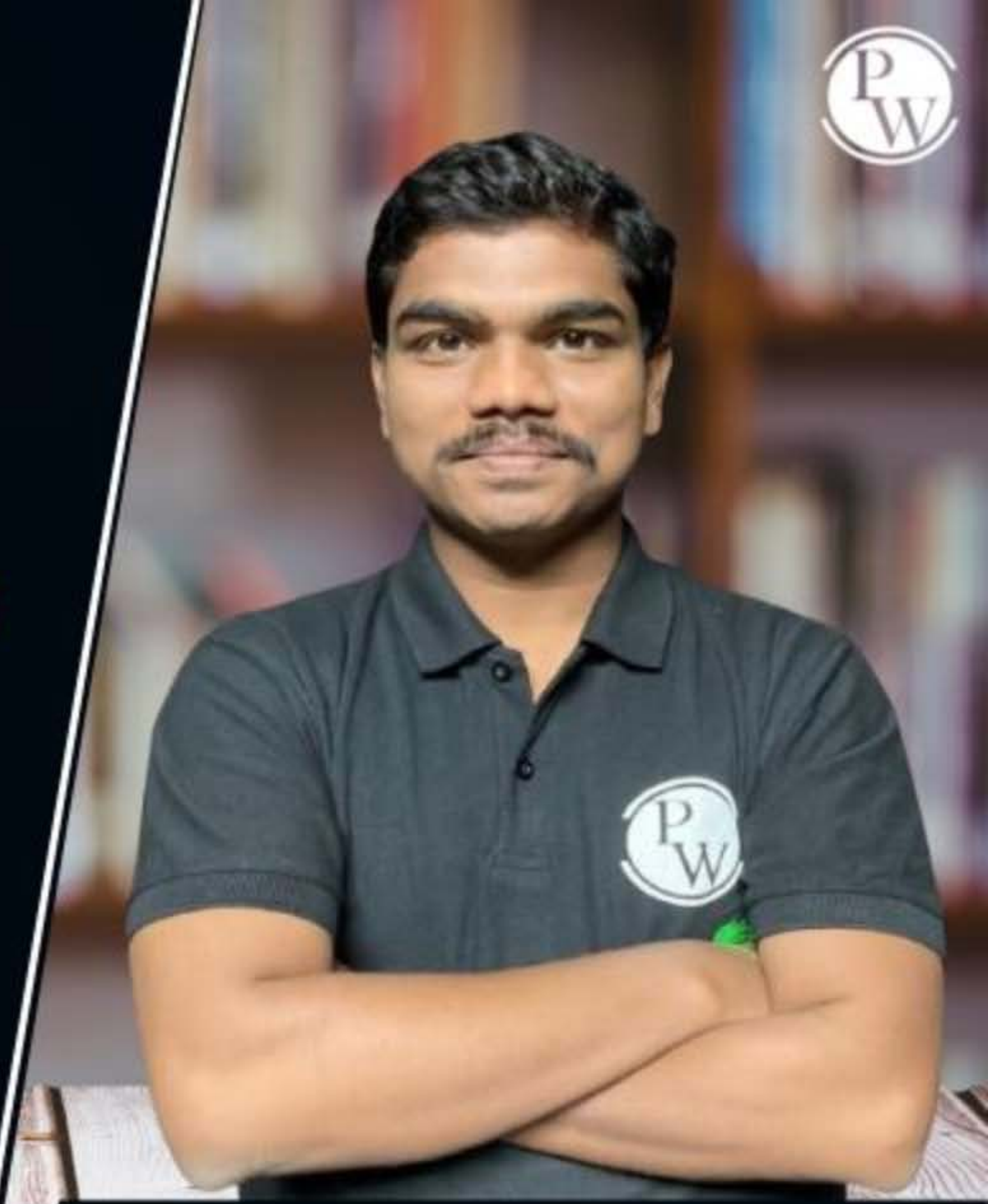
GATE- ALL BRANCHES
ENGINEERING MATHEMATICS

Linear Algebra

Lecture No. 2



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Recap of previous lecture

→ Introduction to Matrices

- order
- Principal diagonal
- trace

→ Types of Matrices

Upper Triangular: $a_{ij} = 0 \ \forall \ i > j$

Lower Triangular: $a_{ij} = 0 \ \forall \ i < j$

→ Diagonal Matrix: $a_{ij} = 0 \ \forall \ i \neq j$

→ Transpose.

→ Symmetric → $A = A^T$.

→ Skew-symmetric → $A = -A^T$.

- → All Principal diagonal elements are zeroes.
- → $tr(\text{skew-symmetric}) = 0$.

→ $$A_{n \times n} = \underbrace{\frac{1}{2}(A + A^T)}_{\text{Symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew-symmetric}}$$

Topics to be Covered

Topic Types of Matrices (contd.)

Topic Matrix Algebra

Matrix Algebra

→ Multiplication of Matrices:

- Two Matrices $A_{m\times n}$ and $B_{p\times q}$ are compatiable for Multiplication if $n = p$. and (AB) is of order $m\times q$.

Let $A_{2\times 3}$ & $B_{3\times 1}$ $\Rightarrow A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2\times 3}$; $B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$$\Rightarrow (AB)_{2\times 1} = \begin{bmatrix} a_{11}b_1 + a_{12}b_2 + a_{13}b_3 \\ a_{21}b_1 + a_{22}b_2 + a_{23}b_3 \end{bmatrix}$$

$(A)_{m\times n}$ & $(B)_{n\times q}$.

$(AB)_{m\times q}$.

then $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}\, b_{kj}$

Let $A_{3\times 4} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3\times 4}$; $B_{4\times 3} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \\ b_{41} & b_{42} & b_{43} \end{bmatrix}_{4\times 3}$

$(AB)_{3\times 3} \rightarrow$ 2nd row 1st column element.

$$(AB)_{21} = a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} + a_{24}b_{41}$$

$$= \sum_{k=1}^{4} a_{2k} \cdot b_{k1}$$

→ Matrix Multiplication is $\boxed{\text{NOT}}$ Commutative.

i.e AB May not be equal to BA.

Ex: Let $A_{3\times 2}$ & $B_{2\times 4}$ be two Matrices.

$$AB = (A)_{3\times 2}(B)_{2\times 4} = (AB)_{3\times 4}$$

Pre factor. Post factor.

For $BA = (B)_{2\times 4}(A)_{3\times 2} = \nexists$.

⊛ → Matrix Multiplication is Associative.

$A(BC) = (AB)C$ Provided all Multiplications are feasible.

→ Two Multiply two Matrices $A_{m\times n}$ & $B_{n\times p}$;

(i) No. of Multiplications needed to calculate AB is $\underline{(mp)n} = mnp$.

(ii) No. of Additions needed to calculate AB is $\underline{(mp)(n-1)} = m(n-1)p$.

→ To Calculate AB Where A is a 3×2 matrix & B is a 2×4 Matrix, No. of Multiplications Needed is $\underline{3\times 2\times 4} = 24$

In general $A_{m\times n}$ & $B_{n\times p}$; $\Rightarrow (AB)_{m\times p}$.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} ; \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & \cdots & b_{1p} \\ b_{21} & b_{22} & b_{23} & \cdots & b_{2p} \\ b_{31} & b_{32} & b_{33} & \cdots & b_{3p} \\ \vdots & \vdots & \vdots & & \vdots \\ b_{n1} & b_{n2} & b_{n3} & \cdots & b_{np} \end{bmatrix}$$

$$(AB)_{11} = \underbrace{a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} + \cdots + a_{1n}b_{n1}}_{\text{To add these 'n' terms}} \longrightarrow \text{'n' multiplications}$$

No. of additions required $= (n-1)$

④ → The Minimum No. of Multiplications required to calculate the Product ABC where A, B, C are Matrices of size 6×5, 5×4, 4×7 respectively is 288.

Sol: $A_{6\times5}$; $B_{5\times4}$; $C_{4\times7}$.

(ABC)

$(AB) \to 6\times5\times4 = 120$

$((AB)C)$: (AB) is 6×4 [120], $C_{4\times7}$ ✓

$\underbrace{6\times4 \;\; 4\times7}$

$6\times4\times7 = 168$

$\Rightarrow \boxed{\text{Total} = 288}$

$A_{6\times5}(BC)$: (BC) is 5×7 [140]

$\underbrace{6\times5 \;\; 5\times7}$

$6\times5\times7 = 210$ ✗

$\boxed{\text{Total} = 350}$

$B_{5\times4}$ $C_{4\times7}$.

$(BC)_{5\times7}$

$5\times4\times7$

$20\times7 = 140$.

$\Rightarrow (AB)^T = B^T A^T.$

in General $\boxed{(A_1 A_2 A_3 \ldots A_n)^T = (A_n)^T \cdot (A_{n-1})^T \cdots (A_3)^T \cdot (A_2)^T \cdot (A_1)^T}$

let $C = (AB) \Rightarrow c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$

$C^T \Rightarrow c_{ji} = \sum_{k=1}^{n} a_{jk} b_{ki} = B^T \cdot A^T.$

$\longrightarrow$ If A, B are two Matrices Such that AB & BA both Exist;

then $\mathrm{tr}(AB) = \mathrm{tr}(BA).$

$A_{3\times4}\ B_{4\times3} \Rightarrow AB \rightarrow 3\times3$

$BA \rightarrow 4\times4$

In general $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$

④ $\Rightarrow$ If A & B are Symmetric Matrices, then

(i) $AB + BA$ is always Symmetric.

Given $A = A^T$ & $B = B^T$.

Let $C = AB + BA$

$$\Rightarrow C^T = (AB + BA)^T = (AB)^T + (BA)^T$$
$$= B^T A^T + A^T B^T$$
$$= BA + AB = AB + BA = C$$

$\Rightarrow$ $\boxed{C = AB + BA \text{ is symmetric.}}$

(ii) $AB - BA$ is always skew-Symmetric.

For $C = AB - BA$.

$$\Rightarrow C^T = (AB - BA)^T = (AB)^T - (BA)^T$$
$$= B^T A^T - A^T B^T = BA - AB = -C$$

$\rightarrow$ The Product of two Non-zero Matrices can be a Zero/Null Matrix.

For $A = \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix}$; $B = \begin{bmatrix} 0 & 0 \\ 0 & b \end{bmatrix}$; $a \neq 0$; $b \neq 0$.

$\Rightarrow AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow$ Null Matrix.

<u>Real Numbers</u>

$xy = 0 \Rightarrow x = 0$ (or) $y = 0$

→ Let $\vec{x}$ be a vector 'A' is a Matrix

$$A_{n\times n}\,\vec{x}_{n\times 1} = \vec{y}_{n\times 1}$$

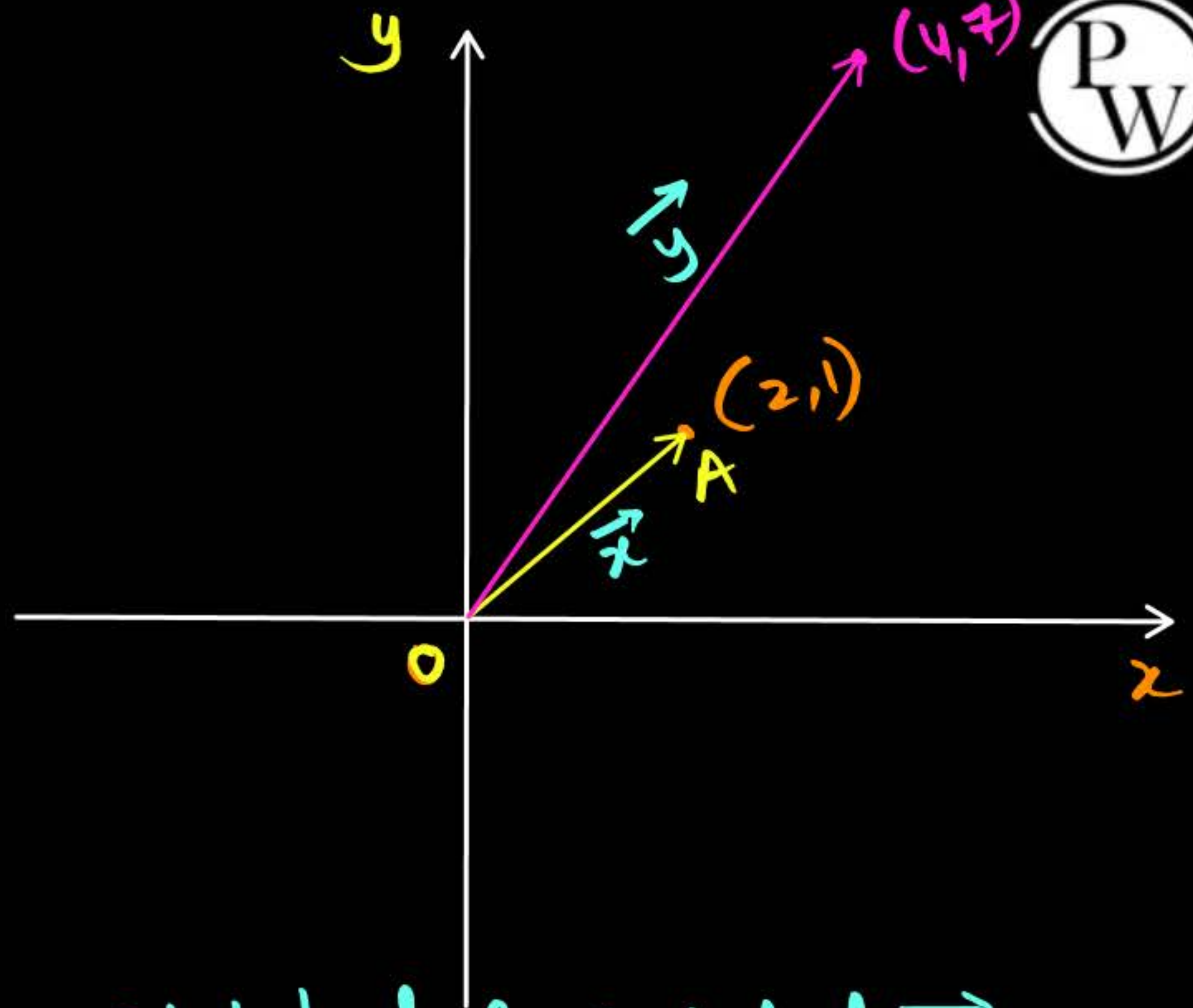
For $n=2$; $A_{2\times 2}\,\vec{x}_{2\times 1} = \vec{y}_{2\times 1}$.

$\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Let $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$

$$\Rightarrow A\vec{x} = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

; $A \rightarrow$ Stetched & rotated $\vec{x}$.

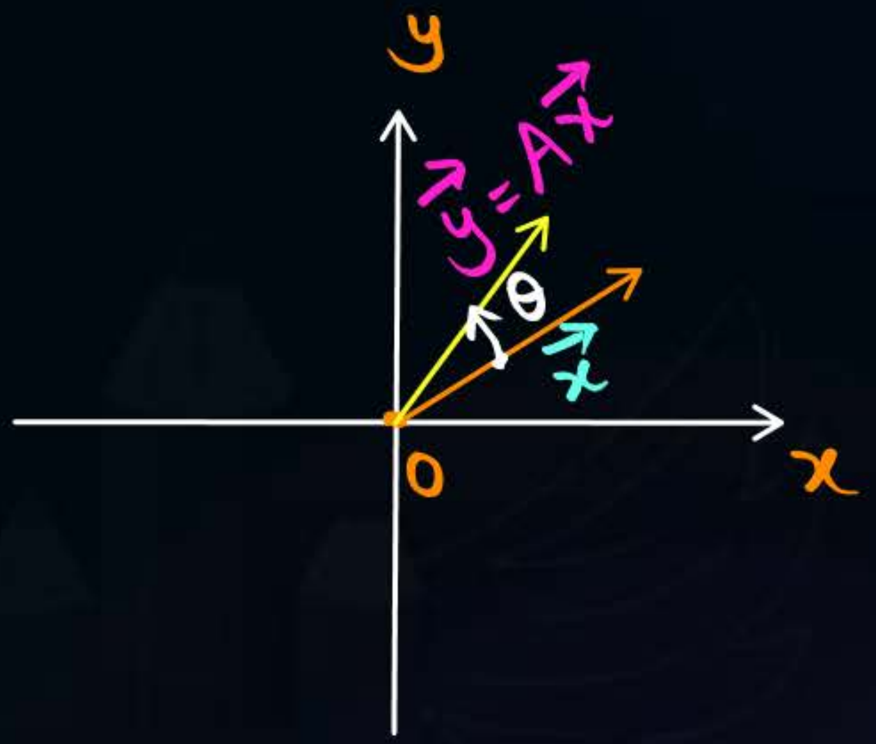


Types of Matrices (contd.)

→ Orthogonal Matrix: A Matrix that Preserves the length of the vector when operated by the Matrix.

$|\vec{x}| \rightarrow$ Magnitude of vector

$\Rightarrow \quad A\vec{x} = |\vec{y}| = |\vec{x}|.$



i.e The Matrix Performs Pure Rotation of $\vec{x}$.

$$\Rightarrow \quad A\vec{x} = \vec{y} \Rightarrow (A^T A)\vec{x} = (A^T y).$$

Ex: $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$ Rotates $\vec{x}$ by 'θ' in Anti-clock sense.

consider $A^T = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\Rightarrow A \cdot A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$

$\Rightarrow \boxed{A \cdot A^T = A^T \cdot A = I}$

$\Rightarrow \boxed{A^T = A^{-1}}$

⊛ → All rows/columns of an orthogonal Matrix are Pairwise orthogonal

i.e $(i^{th} \text{ row}) \cdot (j^{th} \text{ row}) = 0 \ \forall \ i \neq j$

→ Involutary Matrix:

A Matrix 'A' is said to be Involutary if

$$A^2 = I.$$

i.e The Inverse of the Matrix is the Matrix itself.

Ex: $A = \begin{bmatrix} 4 & -1 \\ 15 & -4 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 4 & -1 \\ 15 & -4 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 15 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Since $A^2 = I \Rightarrow A^4 = (A^2)^2 = I \Rightarrow A^6 = A^8 = I.$

i.e Matrix doesn't show any affect when applied even No. of times.

→ Idempotent Matix:

A Matix 'A' is said to be an Idempotent Matix if

$$A^2 = A.$$

(i.e Matrix is operational only once).

Ex: ① Projection Matix.

② $A = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}$

$= \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix} = A.$

$A^3 = A^2 A = A \cdot A = A^2 = A; \Rightarrow A^3 = A \Rightarrow A^4 = A^5 = \cdots = A.$

→ Nilpotent Matrix:

A Non-zero Matrix 'A' is said to be Nil-Potent if $\exists$ 'n' $\in \mathbb{Z}^+$

$\Rightarrow A^n = O$.

The least value of 'n' for which $A^n = O$ is called "Index of the Matrix".

Ex: $A = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix} \neq O$

$\Rightarrow A^2 = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$

$\Rightarrow$ Index = 2

$A^3 = O \Rightarrow A^4 = A^3 \cdot A = O$

$A^5 = A^3 \cdot A^2 = O$

$A^6 = (A^3)^2 = O$

$\Rightarrow A^7 = A^6 \cdot A = O$

→ The Number of 3×3 Symmetric Matrices that can be formed using the digits (1, 2, 3, ..., 9) with repetition allowed is 531441

Sol:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ \cancel{a_{21}} & a_{22} & a_{23} \\ \cancel{a_{31}} & \cancel{a_{32}} & a_{33} \end{bmatrix} \begin{matrix} \longrightarrow \text{No. of choices available} = 3 \\ \longrightarrow \text{No. of " "} = 2 \\ \longrightarrow \text{No. of " "} = 1. \end{matrix}$$

⇒ Total No. of choices available = 6.

Options available for each choice = 9.

⇒ Total No. of Matrices Possible $= 9^6 = 531441$

Note: The Number of $n \times n$ Symmetic Matrices that can be formed using the digits $(1, 2, 3, \ldots, 9)$ is $9^{\frac{n^2+n}{2}}$

For $n = 3$; $9^6 = 531441$

2 mins Summary

→ Types of Matrices

→ Algebra of Matrices.

THANK - YOU